

## Logic PhD Exam, August 2006

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*Please complete seven problems among the following, including one from each section.*

### 1. General logic

1. State the incompleteness theorem and explain the role of Gödel numbering in its proof.
2. Sketch the proof of the compactness theorem for first order logic.
3. Sketch the proof of Tarski's undefinability of truth.

### 2. Model theory

4. Prove that the theory of dense linear orders without endpoints is complete.
5. Find two models which are elementarily equivalent but not isomorphic.
6. State and prove Tarski's criterion for elementary submodels.

### 3. Set theory

7. Sketch the proof of  $L$  satisfying CH.
8. State and prove the infinite Ramsey theorem.
9. Prove that every stationary subset of  $\omega_1$  can be split into  $\aleph_1$  many pairwise disjoint stationary subsets.

### 4. Computability

10. Show how to construct a simple recursively enumerable set  $A$  (meaning that  $\omega - A$  has no infinite r.e. subset).
11. Define the recursively enumerable set  $K = \{a : \{a\}(a) \downarrow\}$ . Use the  $S_1^1$  function to show that  $K$  is many-one complete.
12. Prove that the following are equivalent, for any nonempty  $A \subset \omega$ :
  - (a)  $A$  is the domain of some partial recursive function  $\phi$ .
  - (b)  $A$  is the range of some (total) recursive function  $f$ .